

Evidence Summary - Technology #05

Phage Anti-Biofilm Coating for Shunt Infection Prevention

External Evidence Sources (5 total):

Source 1: Preventing *S. aureus* biofilm formation on titanium surfaces by

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6693497>

Excerpt: by ALR López · 2019 · Cited by 72 — β -peptide-containing films fabricated on rough titanium substrate surfaces can release sufficient quantities of β -peptide to prevent *S. aureus* biofilm ...

Source 2: *Staphylococcus aureus* resistance on titanium coated with

URL: <https://www.sciencedirect.com/science/article/abs/pii/S0142961210010537>

Excerpt: by X Khoo · 2010 · Cited by 79 — TBP4-PEG forms a serum-resistant surface coating capable of preventing *S. aureus* colonization and subsequent biofilm formation.

Source 3: Recent developments in preventing catheter-related infections

URL: <https://pubs.aip.org/aip/bmf/article/18/5/051506/3316469/Recent-developments-in-preventing-catheter-related>

Excerpt: Oct 11, 2024 — Under static conditions, PU catheters coated with oil-infused PDMS surface layers inhibited *E. coli* and *S. aureus* biofilm formation.

Source 4: Antibiotic-Free Solutions for the Development of Biofilm

URL: https://link.springer.com/chapter/10.1007/978-3-031-04484-7_21

Excerpt: by B Costa · 2022 · Cited by 1 — All coatings showed antibacterial and antifungal properties, *Staphylococcus aureus* (MRSA)

Source 5: Bacteriophage-Mediated Control of Biofilm: A Promising

URL: <https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2022.825828/full>

Excerpt: by C Chang · 2022 · Cited by 177 — catheter -associated biofilm clearance may involve phage is used as a gel-like coating on the catheter to reduce the bacterial adhesion to the ...

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*